

LOCKDOWN LAB

Multi-Stage Intrusion Analysis Report

Network Forensics | Memory Analysis | Malware Attribution

Lab Overview

This lab involved reconstructing a multi-stage intrusion by analyzing network traffic, memory dumps, and malware artifacts. Tools used included Wireshark, Volatility, and VirusTotal. All findings were mapped to the MITRE ATT&CK framework.

Tools Used: Wireshark, Volatility, VirusTotal

Framework: MITRE ATT&CK

Malware Family Identified: RAT Agent Tesla

Packer Used: UPX

Step 1 — Conversation Tab: Network Mapping

Objective

Use the Conversation Tab in Wireshark to identify communication between IPs and understand who is talking to whom.

Actions Taken

- Opened the Conversation Tab to view sessions between 2 IPs or ports.
- Investigated: who talked to who, how much data was exchanged, and how active the communication was.
- Noticed Address A (10.0.2.4) sent approximately 4,166 packets to 10.0.2.15 — the IIS (Internet Information Services) server.

IP Role Identification

- 10.0.2.15 → Victim / IIS Server
- 10.0.2.4 → Attacker / Scanner

Finding

The attacker at 10.0.2.4 was actively scanning and communicating with the victim IIS server at 10.0.2.15, generating over 4,166 packets of traffic.

Ethernet · 12	IPv4 · 16	IPv6 · 4	TCP · 1085	UDP · 39								
Address A	Address B	Packets ▾	Bytes	Stream ID	Packets A → B	Bytes A → B	Packets B → A	Bytes B → A	Rel Start	Duration	Bits/s A → B	Bits/s B → A
10.0.2.4	10.0.2.15	6,007	4 MB	6	4,166	3 MB	1,841	184 kB	78.082576	1434.6036	19 kbps	1028 bits/s

Step 2 — Attacker Behavior: TCP Port Analysis & MITRE Mapping

Objective

Analyze the attacker's behavior by reviewing TCP port usage and map observed activity to MITRE ATT&CK techniques.

Actions Taken

- Identified TCP Port 445 (SMB) as the primary target port.
- Filtered: IP.src == 10.0.2.4 && IP.dst == 10.0.2.15 && small packets, to isolate attacker traffic.
- Observed attacker trying to access shared folders on the victim machine.

MITRE ATT&CK Techniques Identified

- T1046 — Network Service Discovery: Attacker discovering network services running on systems.
- T1059 — PowerShell Execution: Command-line execution via PowerShell.
- T1003 — Credential Dumping: Attacker extracting stored credentials.
- T1566 — Phishing: Social engineering used to deliver the payload.
- T1021 — Remote Desktop Use: Attacker leveraging RDP for lateral movement.

Finding

The attacker used SMB (port 445) to probe shared folders and executed multiple MITRE-mapped techniques including credential dumping, phishing, PowerShell execution, and remote desktop access.

Address A	Port A	Address B	Port B
10.0.2.4	663	10.0.2.15	445

Network Service Discovery

Adversaries may attempt to get a listing of services running on remote hosts and local network infrastructure devices, including those that may be vulnerable to remote software exploitation. Common methods to acquire this information include port, vulnerability, and/or wordlist scans using tools that are brought onto a system.^[1]

Within cloud environments, adversaries may attempt to discover services running on other cloud hosts. Additionally, if the cloud environment is connected to a on-premises environment, adversaries may be able to identify services running on non-cloud systems as well.

Within macOS environments, adversaries may use the native Bonjour application to discover services running on other macOS hosts within a network. The Bonjour mDNSResponder daemon automatically registers and advertises a host's registered services on the network. For example, adversaries can use a mDNS query (such as `dns-sd -B _ssh._tcp .`) to find other systems broadcasting the ssh service.^{[2][3]}

ID: T1046

Sub-techniques: No sub-techniques

① **Tactic:** [Discovery](#)

① **Platforms:** Containers, IaaS, Linux, Network Devices, Windows, macOS

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[Version](#) [Permalink](#)

Step 3 — Tree Connect Request & Payload Upload

Objective

Investigate the SMB Tree Connect request to identify what the attacker attempted to access and what malicious payload was uploaded.

Actions Taken

- Located a Tree Connect Request in the capture: `\\10.0.2.15\Documents` and `\\10.0.2.15\IPC$`
- The attacker planted a web-accessible payload designed to grant remote code execution (RCE).

Payload Details

File Name: Shell.aspx

File Size: 101,502 bytes

Finding

The attacker uploaded Shell.aspx (a web shell) to the IIS server via SMB. This file grants remote code execution through the web interface.

2629	240.778552	10.0.2.4	10.0.2.15	SMB2	162 Tree Connect Request, Tree: '\\10.0.2.15\IPC\$'
2630	240.779856	10.0.2.15	10.0.2.4	SMB2	138 Tree Connect Response, Tree: '\\10.0.2.15\IPC\$'
2678	263.312045	10.0.2.4	10.0.2.15	SMB2	172 Tree Connect Request, Tree: '\\10.0.2.15\Documents'
2679	263.316785	10.0.2.15	10.0.2.4	SMB2	138 Tree Connect Response, Tree: '\\10.0.2.15\Documents'

Step 4 — Filtering Reverse Traffic & Port Discovery

Objective

Filter for reverse traffic from victim to attacker to identify callback ports used by the malware.

Actions Taken

- Filtered: IP.src == 10.0.2.15 && IP.dst == 10.0.2.4 && TCP
- Scrolled through filtered traffic and identified Port 4443 being actively used.

Finding

After reversing the filter direction, port 4443 was discovered as the callback/reverse shell port — the channel used by the attacker to receive remote access from the victim machine.

2779	337.992064	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
2783	342.482279	10.0.2.15	10.0.2.4	SMB2	198 Create Request, File: shell.aspx
2784	342.501269	10.0.2.15	10.0.2.4	SMB2	210 Create Response, File: shell.aspx
3505	342.611215	10.0.2.4	10.0.2.15	SMB2	494 Write Request Len:1015024 Off:0, File: shell.aspx
4168	720.000350	10.0.2.15	10.0.2.4	TCP	210 49688 → 4443 [PSH, ACK] Seq=18048 Ack=259169 Win=2101248 Len=176
4161	722.833906	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4164	727.835340	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4167	732.810826	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4170	737.890752	10.0.2.15	10.0.2.4	TCP	54 445 → 37338 [ACK] Seq=11201 Ack=1025486 Win=2100992 Len=0
4171	737.892306	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4174	742.845620	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4177	747.847646	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4180	752.798155	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4183	757.849277	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4186	762.902013	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4189	767.853496	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4192	772.818760	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4195	777.870298	10.0.2.15	10.0.2.4	SMB2	126 KeepAlive Response
4198	780.422735	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18224 Ack=259266 Win=2101248 Len=0
4199	780.465162	10.0.2.15	10.0.2.4	TCP	184 49688 → 4443 [PSH, ACK] Seq=18224 Ack=259266 Win=2101248 Len=130
4202	781.137108	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18354 Ack=259351 Win=2100992 Len=0
4203	781.168798	10.0.2.15	10.0.2.4	TCP	245 49688 → 4443 [PSH, ACK] Seq=18354 Ack=259351 Win=2100992 Len=191
4215	781.272072	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18545 Ack=273951 Win=2102272 Len=0
4236	781.273617	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18545 Ack=303151 Win=2102272 Len=0
4252	781.275247	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18545 Ack=325051 Win=2102272 Len=0
4262	781.275943	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18545 Ack=338191 Win=2102272 Len=0
4272	781.277846	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18545 Ack=350108 Win=2102272 Len=0
4273	781.784271	10.0.2.15	10.0.2.4	TCP	187 49688 → 4443 [PSH, ACK] Seq=18545 Ack=350108 Win=2102272 Len=133
4276	781.805687	10.0.2.15	10.0.2.4	TCP	54 49688 → 4443 [ACK] Seq=18678 Ack=350193 Win=2102272 Len=0
4277	781.856911	10.0.2.15	10.0.2.4	TCP	187 49688 → 4443 [PSH, ACK] Seq=18678 Ack=350193 Win=2102272 Len=133

Step 5 — Understanding Kernel Base (Volatility Prerequisite)

Objective

Before performing memory analysis with Volatility, establish a baseline understanding of the kernel base address.

What is Kernel Base?

- The starting virtual address in memory where the OS kernel is loaded.
- Any kernel symbol or structure is located at: Kernel Base + Offset
- Knowing the kernel base is essential for memory forensics and incident response.

Actions Taken

Ran Volatility against the memory dump to find the kernel base:

```
python vol.py -f memory.raw windows.info
```

Kernel Base Found: 0xF800 7921 3000

To check kernel base using Volatility3:

```
py (or file path of Volatility) -m Volatility3 -f [file] windows.info
```

Finding

Kernel base was identified at 0xF800 7921 3000. This address is essential for correctly interpreting all subsequent memory forensics results.

```
Variable      Value
Kernel Base   0xF80079213000
DTB           0x1aa000
Symbols File:  C:/Users/josea/AppData/Local/Programs/Python/Python314/Lib/site-packages/volatility3/symbols/windows/ntkrnlmp.pdb/EF9A48AFA50FF07C616585BB01919536-1.json.xz
Is64Bit       True
ISPAE         False
Layer.name    0 windowsIntel32e
memory_layer  1 FileLayer
KdVersionBlock 0xF80079613f10
Major/Minor   15.17763
MachineType   34404
KeNumberProcessors 4
SystemTime    2024-09-10 06:14:13+00:00
NtSystemRoot  C:\Windows
NtProductType NtProductServer
NTMajorVersion 10
NTMinorVersion 0
PE MajorOperatingSystemVersion 10
PE MinorOperatingSystemVersion 0
PE Machine    34404
PE TimeDateStamp Sun Nov 10 07:20:39 2075
```

Step 6 — Identifying the Malicious Process (pslist)

Objective

Use Volatility to list running processes and identify any suspicious executables launched by the attacker.

Actions Taken

- A trusted service launched an unfamiliar executable.
- Used pslist to show all running executables:
 - `py -m vol.py -f memdump.mem windows.pslist`
- Changed the command to: windows.cmdline to view command-line arguments.
- Found PID 4332, PPID 2452 — executed w3wp.exe, then launched updatenow.exe

Finding updatenow.exe Path

Located the full path of updatenow.exe:

```
C:\Program Data\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe
```

- MITRE T1547 — Boot or Logon Executable: The malware placed itself in the Startup folder for persistence.

Finding

The attacker's process chain: w3wp.exe (IIS worker) spawned updatenow.exe, which was placed in the Windows Startup folder — ensuring persistence across reboots (T1547).

208	676	svchost.exe	0xcce0970ca080	0	-	0	False	2024-09-10 05:44:45.000000 UTC	N/A	Disabled
4332	2452	w3wp.exe	0xce06574ca080	0	-	0	False	2024-09-10 05:44:45.000000 UTC	2024-09-10 06:10:48.000000 UTC	Disabled
1676	1628	taskhostw.exe	0xce0656e46080	6	-	1	False	2024-09-10 05:52:43.000000 UTC	N/A	Disabled
900	4332	updatenow.exe	0xce0657ddb1c0	3	-	0	True	2024-09-10 06:08:23.000000 UTC	N/A	Disabled

4332	w3wp.exe	-
1676	taskhostw.exe	taskhostw.exe
900	updatenow.exe	"C:\ProgramData\Microsoft\Windows\Start Menu\Programs\Startup\updatenow.exe"

Step 7 — Reverse Shell Analysis: Attack Chain Reconstruction

Objective

Identify how Windows built-in utilities were abused to establish a reverse shell and gain remote command access.

Attack Chain

- Attacker exploits IIS/web server vulnerability.
- w3wp.exe becomes abused as the initial foothold.
- w3wp.exe launches updatenow.exe
- updatenow.exe creates a reverse shell back to the attacker.
- Attacker gains remote command access to the victim machine.

Finding

The attacker used a built-in Windows process (w3wp.exe) to launch a reverse shell via updatenow.exe, achieving full remote command-and-control access — all initiated from the uploaded Shell.aspx web shell.

Step 8 — Packer Identification

Objective

Identify the packer used to obfuscate the malware binary.

Finding

The packer used was UPX — a common executable packer used by attackers to compress and obfuscate malware binaries, making static analysis and signature detection more difficult.

```
PE32
Operation system: Windows (XP) [I386, 32-bit, GUI]
Format: Autolt (3.XX)
Linker: Microsoft Linker (12.00.40629)
Compiler: Microsoft Visual C/C++ (18.00.40629) [POGO_O_CPP]
Language: C++
Tool: Microsoft Visual Studio (2013)
(Heur) Protection: Generic [Section #0 ("UPX0") has RWX + IAT directory empty]
Packer: UPX (3.91) [NRV, brute]
(Heur) Packer: Generic [EntryPoint + Structure and IAT like UPX + Pushal at EP + Imports like UPX (v2.90-3.XX) + Sections like UPX + Sections collision ("UPX") + Section #1 ("UPX1") compressed + Section #0 ("UPX0") has wrong size + Section #0 ("U...
```

Step 9 — FQDN Lookup via VirusTotal

Objective

Identify the Fully Qualified Domain Name (FQDN) that the malware contacts for command-and-control communication.

Actions Taken

- Used VirusTotal to investigate the malware sample and its network indicators.
- VirusTotal is an IOC (Indicator of Compromise) aggregator that helps analysts understand where malicious intent was directed.

Finding

VirusTotal was used to identify the FQDN contacted by the malware, revealing the C2 infrastructure. VirusTotal aggregates IOCs to help analysts understand the full scope of malicious intent and attribution.

Contacted Domains (6) ⓘ			
Domain	Detections	Created	Registrar
bg.microsoft.map.fastly.net	0 / 92	2011-04-18	MarkMonitor Inc.
cp8nl.hyperhost.ua	2 / 92	-	ua.ukrnames
crt.sectigo.com	0 / 92	2018-08-16	-
microsoft.com	0 / 92	1991-05-02	-
sectigo.com	0 / 92	2018-08-16	-
www.microsoft.com	0 / 92	1991-05-02	-

Step 10 — Malware Family Attribution

Objective

Identify the malware family using social engineering context and VirusTotal findings.

Actions Taken

- Applied social engineering analysis to correlate behavioral indicators.
- Cross-referenced VirusTotal results with known malware families.

Finding

Using social engineering and VirusTotal analysis, the malware sample was attributed to the RAT Agent Tesla malware family — a well-known Remote Access Trojan (RAT) used for credential theft, keylogging, and remote control of victim machines.

Verdict: MAL
Score: 100/100
Threat Name: AgentTesla

Summary of Findings

Step	Finding	Detail	Tool / Ref
1	Network Mapping	10.0.2.4 (attacker) sent 4,166 pkts to 10.0.2.15 (victim IIS)	Wireshark Conversations
2	Attacker Behavior	TCP 445 used; T1046, T1059, T1003, T1566, T1021 identified	MITRE ATT&CK
3	Payload Upload	Shell.aspx (101,502 bytes) uploaded via SMB Tree Connect	Wireshark / SMB
4	Callback Port	Port 4443 identified as reverse shell callback port	Wireshark Filter
5	Kernel Base	Kernel base found at 0xF800 7921 3000	Volatility / windows.info

6	Malicious Process	w3wp.exe (PID 4332) spawned updatenow.exe; placed in Startup (T1547)	Volatility pslist
7	Reverse Shell	updatenow.exe created reverse shell granting attacker RCE	Memory / Process
8	Packer	UPX packer used to obfuscate malware binary	Static Analysis
9	FQDN / C2	VirusTotal used to identify C2 FQDN and IOCs	VirusTotal
10	Attribution	Malware family: RAT Agent Tesla	VirusTotal / Social Eng.

Conclusion

The Lockdown Lab demonstrated a complete multi-stage intrusion lifecycle — from initial network reconnaissance through IIS exploitation, web shell upload, process injection, persistence via startup executables, and ultimately a reverse shell granting the attacker full remote command access.

The malware was identified as RAT Agent Tesla, packed with UPX, and leveraged a chain of trusted Windows processes (w3wp.exe) to avoid detection. Memory forensics via Volatility and network analysis via Wireshark were critical in reconstructing the full attack chain and mapping each behavior to MITRE ATT&CK techniques.